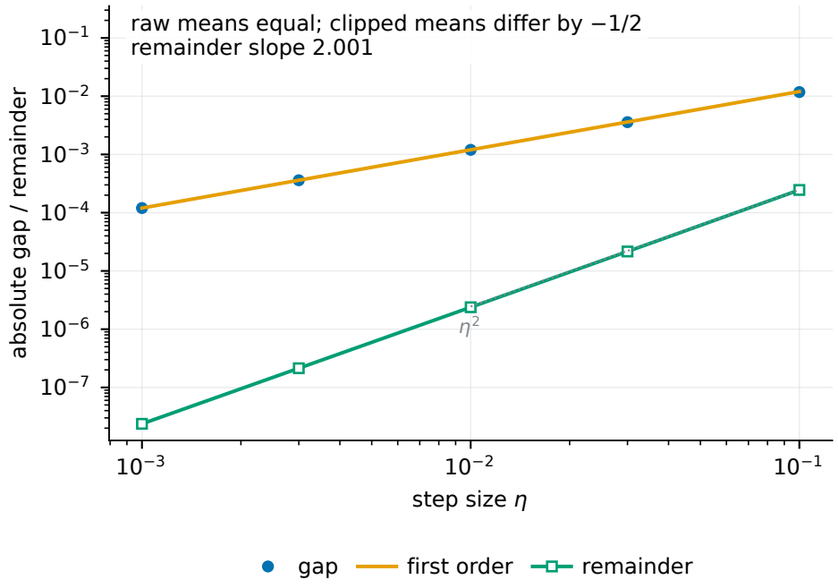
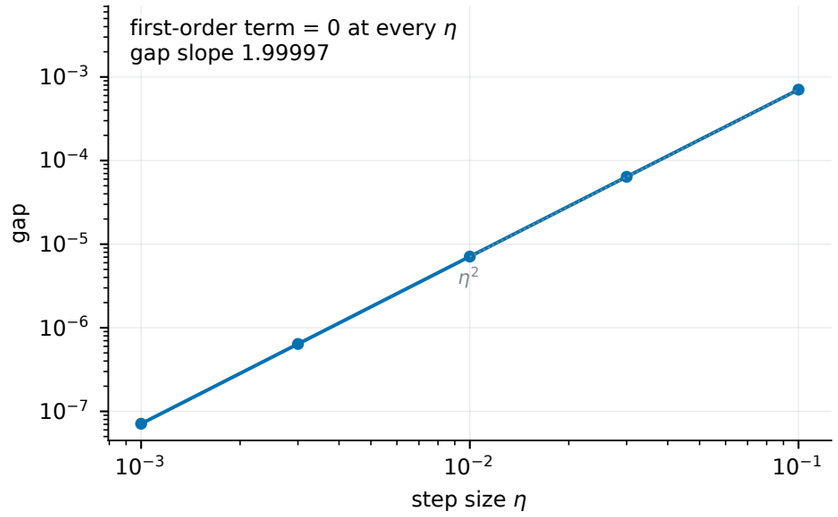
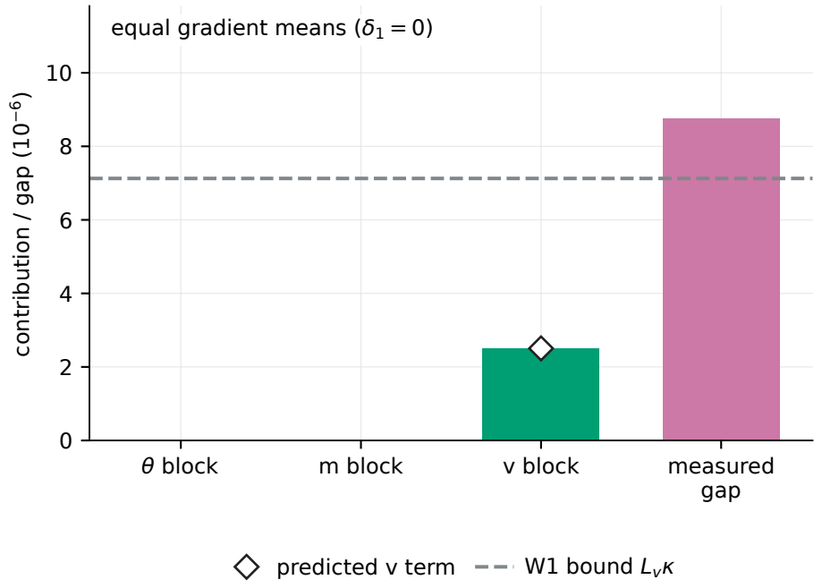
(a) Clipped SGD: the clipped mean decides



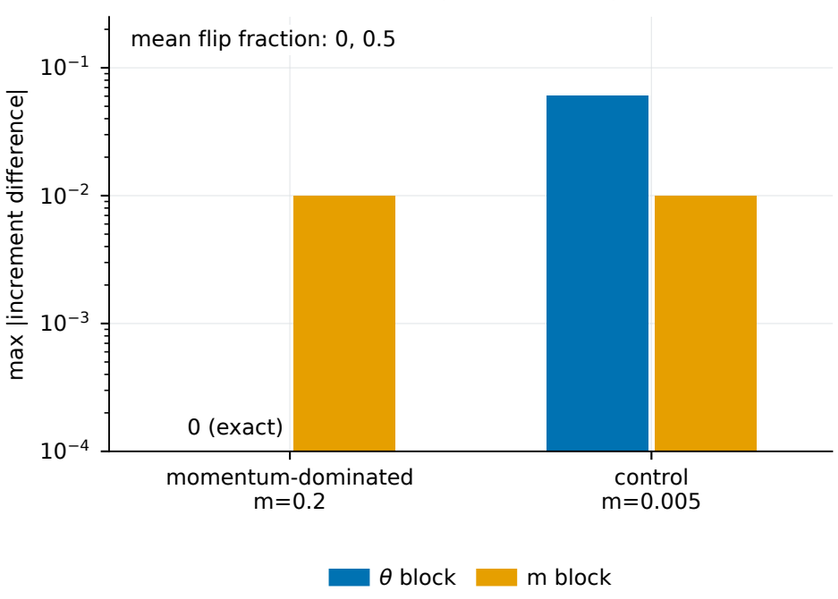
(b) Always-active clipping:
only the direction matters



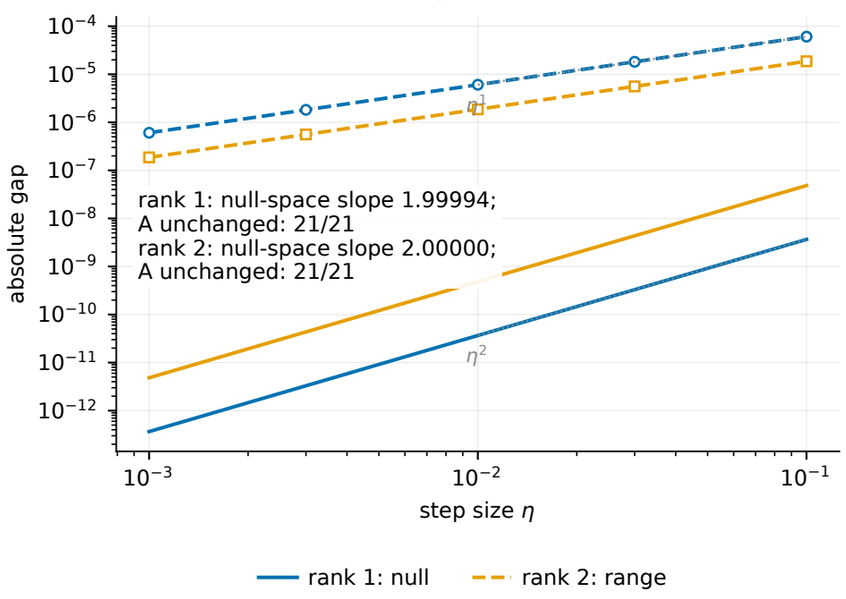
(c) Warm Adam: $E[G^2]$ enters through v



(d) Lion: momentum dominance
blinds the parameter step



(e) LoRA at $B = 0$: only the sketch
$\Gamma A_0^\top$ matters



(f) Learner → first-order statistic

SGD, momentum	EG
warm Adam	EG and EG^2
cold Adam	sign law; EG, EG^2 in m, v
clipped SGD	clipped mean (direction if always active)
Lion	flip probability
LoRA, step 1	sketch $\Gamma A_0^\top$ (or its sign law)

Controlled small networks; not a real-model result. Warm-Adam W1 bound covers only the linear v contribution.